

[YIJC/2025/Q2]

- 2** A teacher has 20 students in her class. The names and dates of birth, in dd-mm-yyyy format, are stored in a list of tuples `students`:

```
students = [('Emma', '12-05-2004'), ('Evelyn', '12-05-2008'),  
            ('Amelia', '14-02-2006'), ('Ava', '14-02-2004'),  
            ('Lucas', '29-08-2003'), ('Benjamin', '19-09-2005'),  
            ('Olivia', '07-01-2003'), ('Grace', '09-12-2003'),  
            ('Ethan', '19-09-2003'), ('Michael', '07-01-2009'),  
            ('Noah', '12-05-2006'), ('James', '22-03-2003')]
```

Task 2.1

Write program code for the function `merge_sort(seq)` using the Merge Sort algorithm to return a new list of the students sorted in ascending order of their names. [6]

Task 2.2

Write program code for the function `sort_dob(seq)` using the function `merge_sort(seq)` written in **Task 2.1** to return a new list of the students sorted in descending order of their ages. [4]